Real 1B: a global signed Rosser-error estimate

Complete parity-dependent error control at fixed inner level

3 November 2026, Hong Kong

Signed-error milestone proved in this research note.
Original prime/composite acceptance unmet. Terminal reserve: 0.

Abstract

At the unchanged inner level $D = X^{199/200}$, we improve the complete Rosser signed-error upper bound from $0.21970750\mathcal{N}_\psi$ to $0.00479332\mathcal{N}_\psi$, up to $o(\mathcal{N}_\psi)$. The improvement comes from selecting a nonnegative majorant according to the actual Möbius sign before applying the output sieve. Its exact guarded first-prime recursion retains every depth. A weighted Mertens calculation evaluates the favorable signed part of its integer comparison, and the previous output-switching mechanism transfers the resulting nonnegative average to the fixed-shift source. All extra roughness boundaries, coprimality diagonals, and tiny first-prime tails are paid. The signed midpoint itself remains unestimated, so no positive twin-prime reserve follows. We assert neither historical novelty nor independent external certification.

1 Source, scope, and the theorem

Fix $\psi \in C_c^\infty((1,2))$, $\psi \ge 0$, $I_\psi = \int \psi > 0$. Set

$$\rho = \frac{8}{31}, \quad z = X^\rho, \quad D = X^\theta, \quad \mathfrak{S}_2 = 2\prod_{p>2}\left(1 - \frac{1}{(p-1)^2}\right), \quad \mathcal{N}_\psi = \mathfrak{S}_2 I_\psi X/\log X.$$

The physical source and its correlation are

$$a_X(n) = \psi(n/X)\Lambda(n+2)\mu^2(n)\mathbf{1}_{P^-(n)>z}, \qquad \mathcal{C}_\mu = \sum_n a_X(n)\mu(n).$$

For sufficiently large X, the source has at most three prime factors. Thus $A = P_1 + P_2 + P_3$ and

$$\mathcal{C}_\mu = -P_1 + P_2 - P_3, \qquad 2P_1 + \mathcal{C}_\mu = A - 2P_3. \tag{1}$$

Let u_D, l_D be the raw upper and lower Rosser divisor sums with cutoff z, and put $M_D = (u_D + l_D)/2$. The precise definitions appear below. We retain the exact decomposition

$$\mathcal{C}_\mu = \mathcal{S}_\theta + \mathcal{R}_\theta, \tag{2}$$

where, with odd n throughout,

$$\mathcal{S}_\theta = \sum_n \psi(n/X)\Lambda(n+2)\mu(n)M_D(n;z),$$

$$\mathcal{R}_\theta = \sum_n \psi(n/X)\Lambda(n+2)\mu(n)(\mathbf{1}_{P^-(n)>z} - M_D(n;z)).$$

The identity uses $\mu^2\mu = \mu$ and does not remove the prime boundary.

Write f, F for the standard linear-sieve functions and define

$$\Delta_\theta(v) = \frac{e^{-\gamma}}{v}(F(\theta/v) - f(\theta/v)), \qquad B_\theta = \frac{1}{2}\Delta_\theta(\rho), \tag{3}$$

$$b_2 = \log(23/8), \qquad b_3 = \int_\rho^{1/3} \frac{\log((1-2t)/t)}{t(1-t)}\,dt, \tag{4}$$

$$\kappa_\mu = -1 + b_2 - b_3 = -0.063036262676\ldots. \tag{5}$$

Theorem 1.1 (Complete signed-error bound)**.** *For fixed $3\rho < \theta < 1$, the stated standard sieve and convolution inputs imply*

$$\boxed{\limsup_{X\to\infty} \frac{\mathcal{R}_\theta}{\mathcal{N}_\psi} \le \frac{5}{3}(B_\theta + \kappa_\mu).} \tag{6}$$

In particular, at the unchanged $\theta = 199/200$,

$$\boxed{\mathcal{C}_\mu \le \mathcal{S}_{199/200} + (0.00479332 + o(1))\mathcal{N}_\psi.} \tag{7}$$

The proof is given in Sections 2–4. Its analytic inputs are the standard upper sieve with factorable remainder, Pascadi's upper-support convolution mechanism, the unbalanced convolution theorem of Fouvry–Radziwiłł, ordinary prime summation, and Mertens cancellation. This is a derived signed-error theorem, not a new theorem about the remaining signed midpoint. The inner level $X^{.995}$ is not a distribution level for primes; the output levels remain $X^{3/5-\varepsilon}$ and $X^{1/2-\varepsilon}$.

Evidence boundary. The Route A/C7 coefficients used in Section 5 are inherited research inputs, not recertified here. The current attachment named `20261102-173831` was unavailable in this request; no new assertion uses its contents. This note instead uses the formulas in the conversation, the preceding delivered gap manuscript, and the primary sources below.

2 Exact sign retention before the output sieve

2.1 Raw weights and the pointwise sandwich

For odd squarefree $d = p_1 \cdots p_r$, arrange the primes in descending order. The upper support $\mathcal{D}^+(D)$ requires $p_1 \cdots p_{j-1} p_j^3 \le D$ at odd j. The lower support $\mathcal{D}^-(D)$ requires $p_1^2 \le D$ and the analogous inequalities at even j. Include 1. Following [1, Definition 5.1], put

$$\lambda_D^{\pm}(d) = \mu(d)\mathbf{1}_{d\in\mathcal{D}^{\pm}(D)}.$$

Let $P(y)$ contain the odd primes at most y, and define

$$u_D(n;y) = \sum_{d\mid(n,P(y))} \lambda_D^+(d), \quad l_D(n;y) = \sum_{d\mid(n,P(y))} \lambda_D^-(d), \quad R_y(n) = \mathbf{1}_{(n,P(y))=1}.$$

Write $\Gamma_D = u_D - l_D$ and $M_D = (u_D + l_D)/2$. For $D \ge y^2$, the pointwise inequalities give $l_D \le R_y \le u_D$. Set

$$r_D(n;y) = \mu(n)(R_y(n) - M_D(n;y)), \qquad H_D(n;y) = \frac{\mu^2(n)}{4}\Gamma_D(n;y) + \frac{1}{2}r_D(n;y). \tag{8}$$

Lemma 2.1 (A nonnegative parity-dependent error majorant)**.** *For all odd inputs, including nonsquarefree ones,*

$$\boxed{0 \le H_D, \qquad r_D \le H_D \le \frac{\mu^2\Gamma_D}{2} \le \frac{\Gamma_D}{2}.} \tag{9}$$

Proof. Write $E^+ = u_D - R_y \ge 0$ and $E^- = R_y - l_D \ge 0$. If $\mu(n) = 1$, then $H_D = E^-/2$; if $\mu(n) = -1$, then $H_D = E^+/2$; if $\mu(n) = 0$, then $H_D = 0$. Equivalently, $|r_D| \le \mu^2\Gamma_D/2$, which gives every inequality in (9). The squarefree factor on Γ_D is essential. □

2.2 The exact recursion covers every depth

If $D > y^3$, the upper and lower support conditions interchange on removing the largest prime of a sieve divisor:

$$\lambda_D^+(pe) = -\lambda_{D/p}^-(e), \qquad \lambda_D^-(pe) = -\lambda_{D/p}^+(e) \quad (P^+(e) < p \le y).$$

The extra lower guard follows from $p^3 < D$. Consequently

$$\begin{aligned} R_y(n) &= 1 - \sum_{p\mid n,\ p\le y} R_{p^-}(n/p), \\ M_D(n;y) &= 1 - \sum_{p\mid n,\ p\le y} M_{D/p}(n/p;p^-), \\ \Gamma_D(n;y) &= \sum_{p\mid n,\ p\le y} \Gamma_{D/p}(n/p;p^-). \end{aligned}$$

Here p^- means that the cutoff excludes p itself. For squarefree n, $\mu(n) = -\mu(n/p)$, so the signed error has the same positive recursion as Γ_D. For general n the exact identity is

$$\boxed{H_D(n;y) = \sum_{p\mid n,\ p\le y} \mathbf{1}_{(p,n/p)=1} H_{D/p}(n/p;p^-).} \tag{10}$$

If a repeated factor is p, its term is killed by the guard. If a different prime repeats, the child H vanishes. This proves (10) on all inputs.

The sum is organized by the first prime of a sieve divisor. It does not discard a stopping term because its *last* prime is small. In fact a bound for all small-last-prime stops by $\Gamma_D(n;X^\eta)$ would be invalid: earlier prefix primes may exceed X^η.

3 Complete transfer to the fixed-shift prime weight

We give the relevant previous adapter and pay the new obligations for H. For odd n, put

$$K(n) = \prod_{\ell\mid n,\ \ell>2} \frac{\ell-1}{\ell-2}, \qquad g_n(q) = \frac{\mathbf{1}_{(n,q)=1}}{\varphi(q)}.$$

The standard upper sieve's one-sided dimension axiom is uniform in n, because $g_n(\ell) \le 1/(\ell-1)$. Its weights, chosen from prime boxes and small-factor sieve weights, do not depend on the local density. Apply the standard singleton upper theorem [2, Remark 2.9, Theorem 2.10, Section 5] to $\{n+2\}$. Its centered remainder is exactly

$$\mathbf{1}_{q\mid n+2} - \frac{\mathbf{1}_{(n,q)=1}}{\varphi(q)}.$$

At output level $X^{h-\varepsilon}$ and cutoff $\sqrt{3X}$ the main term is at most

$$\left(\frac{2}{h-\varepsilon} + O(\epsilon_{\text{sieve}}) + o(1)\right) \frac{\mathfrak{S}_2 K(n)}{\log X}.$$

Multiply by the original nonnegative child in (10) before any Mellin separation. For actual prime outputs, restore the Mangoldt factor $\log X + O(1)$. Proper prime powers have total cost $O(X^{1/2+\epsilon})$, since all input multiplicities are divisor bounded.

3.1 Separation and its errors

Let $Y_0 = X^{1/(\log\log X)^3}$ and let the relative smoothing width be $h_0 = (\log X)^{-B}$. For $p \ge Y_0$, expand the child as

$$H_{D/p}(m;p^-) = \frac{\mu^2(m)}{4}\Gamma_{D/p}(m;p^-) + \frac{\mu(m)}{2}\big(R_{p^-}(m) - M_{D/p}(m;p^-)\big).$$

For a squarefree divisor d of m, let $B_+(d)$ be the maximum of its odd-index Rosser products and $B_-(d)$ the maximum of its even-index products and first-prime square. Put $B_\pm(1) = 1$. The raw support terms are specified by

$$P^+(d) < p, \qquad pB_+(d) \le D, \qquad pB_-(d) \le D.$$

Smooth Mellin inversion separates these scalar tests. The cofactor coefficients remain $O(\tau(m))$, even after multiplication by $\mu(m)$ or $\mu^2(m)$. The prime coefficient is a polylogarithmic Mellin twist of a prime indicator and has the required Siegel–Walfisz property. Only logarithmic powers are lost in kernel norms and partial summation.

The old support-boundary incidences have total

$$E_{\text{support}} \ll X(h_0 + Y_0^{-1})\log X + D\log X.$$

There is one additional scalar test, $R_{p^-}(m) = \mathbf{1}_{p \le P^-(m)}$. Its boundary has $m = qa$ with $q = P^-(m)$ in a relative $O(h_0)$ interval around p. Counting all integer q and all multiples, rather than just the admissible ones, gives

$$\begin{aligned} E_{\text{rough}} &\ll 2X \sum_{p \ge Y_0} \frac{1}{p} \sum_{|q/p-1| \ll h_0} \frac{1}{q} \\ &\ll X(h_0 \log X + Y_0^{-1}). \end{aligned}$$

Only $pq \le 2X$ contributes, and the count of multiples is at most $2X/(pq)$, so no additional large floor error is hidden here. The gcd guard $p \nmid m$ may be removed temporarily during separation: its diagonal $n = p^2a$ has $O(X/Y_0)$ incidences. Restore it in the exact identity.

Every boundary must be paid in the progression remainder as well as in the original Mangoldt sum. If $B(n)$ counts any such incidences, it is bounded by a fixed divisor power. Weighted Cauchy–Schwarz gives

$$\sum_{n \asymp X} B(n)\tau(n+2)^k \le \big(\sum B(n)\big)^{1/2}\big(\sum B(n)\tau(n+2)^{2k}\big)^{1/2}.$$

Ordinary separate divisor moments bound the second factor by $X^{1/2}\log^C X$. The combined error is therefore

$$O\Big(X\sqrt{h_0 + Y_0^{-1}}\log^C X + \sqrt{XD}\log^C X\Big). \tag{11}$$

For fixed $\theta < 1$, choosing B sufficiently large pays every prescribed logarithmic saving. The principal comparison adds only logarithmic powers.

3.2 All first-prime ranges

Fix $0 < \eta < 1/200$. For $X^\eta \le p \le z$, use output level $X^{3/5-\varepsilon}$. The prime factor is shorter than $X^{1/3}$; Pascadi's Propositions 4.4 and 5.2 apply after the separation. The divisor-bounded coefficient reduction is the one used in his Proposition 4.5. This pays the full centered remainder with limiting coefficient 10/3, retaining the coprime comparison.

For $Y_0 \le p < X^\eta$, use output level $X^{1/2-\varepsilon}$ and [3, Corollary 1.1(i)]. Its permitted short-factor range contains

$$\exp((\log X)^{\epsilon_{\mathrm{FR}}}) \le p \le Q^{-11/12} X^{17/36-\epsilon_{\mathrm{FR}}},$$

with $Q \le X^{1/2-\varepsilon}$ and the chosen small η. The resulting limiting coefficient is 4. By (10), this intermediate positive mass is at most $H_D(n; X^\eta) \le \Gamma_D(n; X^\eta)/2$.

The entire tiny-first-prime contribution is $H_D(n; Y_0^-)$. It is at most half the corresponding raw gap. The fundamental lemma with $s = \theta(\log\log X)^3$ and elementary divisor counting give

$$\sum_n \psi(n/X)\Gamma_D(n; Y_0^-) \ll_A X/\log^A X + D.$$

Even the trivial $\Lambda \le \log(3X)$ pays this tail. Thus arbitrary depth is retained; it is not approximated by a finite stopping family.

In particular, if $\langle H\rangle_X$ denotes $(\log X)/(XI_\psi)\sum_n \psi(n/X)K(n)H_D(n;z)$, then the preceding argument gives

$$\limsup \frac{\mathcal{R}_\theta}{\mathcal{N}_\psi} \le \frac{10}{3}\limsup\langle H\rangle_X + \frac{4-10/3}{2}\Delta_\theta(\eta). \tag{12}$$

The last term tends to zero as $\eta \downarrow 0$. Limits are taken with θ, η and all sieve margins fixed first; $X \to \infty$ precedes the removal of output margins and then of η.

4 The favorable integer comparison is proved

4.1 Uniform weighted coprime Mertens cancellation

On odd integers, define a multiplicative function g by

$$g(p^j) = -\frac{1}{p-2} \quad (j \ge 1), \qquad g(1) = 1.$$

Euler factors give the exact convolution

$$\mu K = \mu * g, \qquad \sum_{r\ge 1,\ r \text{ odd}} \frac{|g(r)|}{\sqrt{r}} < \infty. \tag{13}$$

The second statement follows from the Euler factors $1 + 1/((p-2)(\sqrt{p}-1))$, whose increments are $O(p^{-3/2})$.

For fixed $\delta > 0$, $d \le X$, and $M \ge X^\delta$, ordinary uniform coprime Mertens cancellation gives

$$\sum_{(t,2d)=1} \mu(t)\psi(t/T) \ll_{A,\delta} T/\log^A X \qquad (T \ge X^{\delta/2}).$$

One elementary uniformity proof expands the restriction using numbers supported on primes dividing $2d$, splits at $\sqrt{T}$, and applies Rankin with exponent $1/\log\log X$ to the tail. The local product over primes dividing d is at most a fixed logarithmic power; ordinary Mertens cancellation handles the short part. The published uniform integer-parameter version [4, Lemma 3.9(ii)] also suffices.

Apply this to (13), with the exact restrictions $(r,d) = (t,d) = 1$. For $r \le \sqrt{M}$, it pays the inner t-sum, while the absolute sum of $|g(r)|/r$ is bounded. For $r > \sqrt{M}$, trivial counting and (13) give a total $O(M^{3/4})$. Hence, uniformly for $d \le X$ and $M \ge X^\delta$,

$$\sum_{(m,2d)=1} \mu(m)K(m)\psi(m/M) \ll_{A,\delta,\psi} M/\log^A X. \tag{14}$$

Let $\beta_\theta(d) = \frac{1}{2}(\mathbf{1}_{d\in\mathcal{D}^+(D)} + \mathbf{1}_{d\in\mathcal{D}^-(D)})$ on odd squarefree $d \mid P(z)$, and zero otherwise. The exact cofactor expansion, together with coprimality, is

$$\sum_n \psi(n/X)K(n)\mu(n)M_D(n;z) = \sum_d \beta_\theta(d)K(d) \sum_{(m,2d)=1} \mu(m)K(m)\psi(dm/X).$$

Here $M = X/d \geq X^{1-\theta}$ and $K(d)$ costs only a logarithmic power. Summing (14) proves

$$\sum_n \psi(n/X)K(n)\mu(n)M_D(n;z) = O_A(X/\log^A X). \tag{15}$$

This estimate is for an integer comparison, without $\Lambda(n+2)$.

4.2 The raw gap and the rough signed mean

The exact local-density computation is, uniformly for odd squarefree d,

$$\sum_{n \text{ odd},\ d\mid n} \psi(n/X)K(n) = \frac{XI_\psi}{\mathfrak{S}_2\varphi(d)} + O_\psi(\log^3 X).$$

For example expand $K = 1 * h$, where $h(p) = 1/(p-2)$ and h is supported on odd squarefree numbers; counting multiples of $[d,r]$ and its Euler product give this formula. Summing against raw Rosser coefficients costs $O(D\log^3 X) = o_A(X/\log^A X)$ since $\theta < 1$. The linear-sieve main terms therefore yield

$$\sum_n \psi(n/X)K(n)\Gamma_D(n;z) \leq (\Delta_\theta(\rho) + o(1))XI_\psi/\log X. \tag{16}$$

For the ordinary rough integer source, $K(n) = 1 + O_\rho(X^{-\rho})$. Prime summation over its one-, two-, and three-factor pieces gives

$$\sum_n \psi(n/X)K(n)\mu(n)R_z(n) = (\kappa_\mu + o(1))XI_\psi/\log X. \tag{17}$$

No shifted-prime asymptotic is assumed in (17).

Combining (8), (15), (16), and (17), using $\mu^2\Gamma_D \leq \Gamma_D$, gives

$$\limsup\langle H\rangle_X \leq \frac{\Delta_\theta(\rho)}{4} + \frac{\kappa_\mu}{2}.$$

Insert this in (12), let $\eta \downarrow 0$, and obtain

$$\frac{10}{3}\left(\frac{\Delta_\theta(\rho)}{4} + \frac{\kappa_\mu}{2}\right) = \frac{5}{3}(B_\theta + \kappa_\mu).$$

This proves Theorem 1.1, with the full normalization $\mathcal{N}_\psi$.

5 Certified constants and exact terminal implications

For $3\rho < \theta \leq 1$, put $s = \theta/\rho < 4$. The sieve functions give

$$B_\theta = \frac{1}{\theta}\left[1 + \int_3^s \frac{\log(v-2)}{v-1}\,dv - \log(s-1)\right].$$

The substitution $t = 1/v$ shows the exact identity

$$B_1 = 1 + b_3 - b_2 = -\kappa_\mu.$$

This numerical identity does not invoke an analytic theorem at $D = X$. The proof remains at fixed $\theta = 199/200 < 1$. The companion directed-rounding certificate gives

$$0.00479330753294\ldots \le \frac{5}{3}(B_{199/200} - B_1) \le 0.00479330837338\ldots < 0.00479332. \tag{18}$$

Retain the supplied research inputs

$$A \ge (1.02231944989 - o(1))\mathcal{N}_\psi, \qquad P_3 \le (0.396858377821267460 + o(1))\mathcal{N}_\psi.$$

Then (1) has threshold $c_* = 0.228602694247465080$.

At the same $\theta = .995$	Error upper cost	Remaining signed allowance
Previous absolute gap	0.21970750	0.008895194247…
This signed majorant	0.00479332	0.223809374247…

The reduction in the safe error debit is $0.21491418\,\mathcal{N}_\psi$. It is a proved improvement of this comparison bound, not an earned lower bound for prime mass or a measured value of the actual error.

For example, the following additional estimate would suffice:

$$\boxed{\mathcal{S}_{199/200} \le (0.22 + o(1))\mathcal{N}_\psi.} \qquad \textbf{NOT PROVED} \tag{19}$$

It would give $P_1 \ge (0.00190468 - o(1))\mathcal{N}_\psi$ on matched scales. Neither this conditional reserve nor the enlarged allowance is a completed terminal reserve. A numerically wider allowance does not establish that the required signed estimate is easier to prove.

6 The other probes and the remaining arithmetic

6.1 Averaged multipliers: a benchmark and an unpaid gate

A useful benchmark can be proved by classical Bombieri–Vinogradov and Mertens cancellation. For fixed $\delta, \epsilon > 0$, $X^\delta \le M \le X^{1/2-\epsilon}$, $DM = X$, and fixed smooth box weights,

$$\sum_{m \text{ odd}} \mu(m)\eta(m/M) \sum_{d \text{ odd},\ (d,m)=1} \mu^2(d)w(d/D)\psi(dm/X)\Lambda(dm+2) \ll_A X/\log^A X. \tag{20}$$

Here there is no Rosser factorization gate on d. To check the adapter, expand squarefreeness via $r^2 \mid d$ while retaining $(r, m) = 1$, then expand coprimality via $k \mid m$. Truncate $r, k \le (\log X)^B$ with elementary absolute tail estimates. The output moduli are exactly $q = mkr^2 \le M(\log X)^{3B}$, in the BV range. Since m is squarefree and $(r, m) = 1$,

$$\varphi(mkr^2) = k\varphi(m)\varphi(r^2).$$

Extending the main k-sum gives $\sum_{k|m} \mu(k)/k = \varphi(m)/m$, leaving a coprime Mertens sum in m. The truncation and floor errors have fixed power or arbitrary logarithmic savings because both M, D are polynomial in X.

The actual coefficient is $\beta_\theta(d) = \mu^2(d)b_\theta(d)$. After $v = dm + 2$, its progression sum contains $b_\theta((v-2)/m)$, a factorization-dependent weight. Ordinary BV does not cover that weight. The complement $1 - b_\theta$ has no proved favorable sign, and $d = 1$ as well as longer m ranges remain. Thus (20) contributes no quantified saving to the full source.

A determinant second-moment attempt likewise leaves the signed quadratic form with kernel

$$K_\beta(m_1, m_2) = \sum_{d,\ (d, m_1m_2)=1} \beta_\theta(d)w(d/D)\psi(dm_1/X)\psi(dm_2/X)\Lambda(dm_1+2)\Lambda(dm_2+2).$$

Entrywise sieve upper bounds do not bound its quadratic form with coefficients $\mu(m_1)\mu(m_2)$: negative entries in that signed sum reverse the inequality. No quadratic-form domination or favorable signed aggregate has been proved here.

6.2 Precise surviving Vaughan tail

The preceding round already paid the small-modulus Type I terms. For $U = (\log X)^{K_0}$ and $\alpha_U(c) = \sum_{ab=c,\ a,b>U} \mu(a)\Lambda(b)$, it gives

$$\mathcal{S}_\theta = \mathcal{T}_{\theta,U} + O_A(X/\log^A X),$$

$$\mathcal{T}_{\theta,U} = \sum_{d|P(z)} \beta_\theta(d) \sum_{(m,2d)=1} \mu(m)\psi(dm/X) \sum_{c|dm+2} \alpha_U(c). \tag{21}$$

No complete one-sided bound for (21) was obtained in this round. The outer divisor sum, all moduli, proper prime powers, and original prime boundary remain in the identity. Replacing the inner divisor sum by $\alpha_U(dm+2)$ would not be valid.

6.3 Independent literature at its exact scope

The current Lü–Wang–Wu preprint [5] states cancellation of $\mu(p+2)$ over unrestricted primes. Its public statement does not supply the polynomial multiplier family with the Rosser gate and the necessary $X/\log X$ weighted error; no opposite-shift or uniform affine extension is assumed. Ford–Maynard [6] requires actual Type I/II information on the difference sequence and shows why a narrow Type II range can still admit prime-free models. Neither paper is used as an unproved substitute for (21). Recent rough-pair parity statements inspected in a follow-up repository concern subpolynomial roughness and do not force a prime endpoint; they are not inputs to Theorem 1.1.

7 Verification and acceptance

The companion standalone Python program checks the pointwise inequalities, the guarded H recursion including nonsquarefree inputs, and the local convolution (13). Its integer-directed logarithms and midpoint quadrature certify (18). These finite checks supplement the general proofs; they do not certify the external analytic inputs, inherited A/C7 bounds, or the unproved midpoint estimate.

The exact limitation is especially transparent: on every original rough input, $R_z = M_D = 1$ and $\Gamma_D = 0$. Hence

$$r_D(n;z) = H_D(n;z) = 0 \qquad (P^-(n) > z).$$

The new estimate controls the signed error created outside that rough support. It does not itself distinguish primes from semiprimes inside the physical source. That information remains in $\mathcal{S}_\theta$.

What done. A complete one-sided signed-error bound at fixed $\theta = .995$, with safe debit 0.00479332 instead of 0.21970750; all first-prime ranges, coprimality, boundaries, and depths included.

What remains. A favorable estimate for the original-source midpoint/Vaughan tail. No zero-prime exclusion, no positive terminal reserve, and no G5 claim. The stronger requested acceptance remains unmet.

Next frontier. The actual factorization-gated affine correlation, or a joint estimate that controls it with its complement while retaining the prime boundary. The new error budget is quantified, not automatically consumable.

References

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